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Sensor apparatus

Abstract

Provided is a sensor apparatus that can suppress movement of foreign matter from a Peltier element to a sensor element. The sensor apparatus includes a package substrate, a Peltier element, a circuit substrate, and a sensor element. The package substrate has a recess portion on a side of a first surface and plural terminals on a side of a second surface located on an opposite side of the first surface. The Peltier element is arranged in the recess portion. The circuit substrate is arranged on an opposite side of a bottom surface of the recess portion with the Peltier element sandwiched therebetween. The sensor element is attached to an opposite side of a surface, the surface being opposed to the Peltier element.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
(1) This application is a U.S. National Phase of International Patent Application No. PCT/JP2020/004709 filed Feb. 7, 2020, which claims priority benefit of Japanese Patent Application No. JP 2019-068608 filed in the Japan Patent Office on Mar. 29, 2019. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.
TECHNICAL FIELD
(2) The present disclosure relates to a sensor apparatus.
BACKGROUND ART
(3) An airtight sealed package incorporating a Peltier element is known as means of cooling a solid-state imaging apparatus (refer, for example, to PTL 1).
CITATION LIST
Patent Literature
(4) [PTL 1]
(5) Japanese Patent Laid-Open No. 2003-258221
SUMMARY
Technical Problem
(6) In various processes involving handling of a Peltier element such as a process for manufacturing a Peltier element, a process for transferring the Peltier element, and a process for attaching the Peltier element to a package by using solder or other means, there is a possibility that foreign matters such as dust present in a work environment may adhere inside the Peltier element or to its outer surface. There is a possibility that these foreign matters may detach themselves from the Peltier element, move inside the package, and adhere to a sensor element inside the package, after the Peltier element is arranged inside the package. If a foreign matter adheres to the sensor element, there is a possibility that the foreign matter that has adhered to the sensor element may affect operation of the sensor element.
(7) The present disclosure has been devised in light of the foregoing, and it is an object of the present disclosure to provide a sensor apparatus that can suppress the movement of the foreign matter from the Peltier element to the sensor element.

Solution to Problem

(8) A mode of the present disclosure is a sensor apparatus that includes a package substrate having a recess portion on a side of a first surface and plural terminals on a side of a second surface located on an opposite side of the first surface, a Peltier element arranged in the recess portion, a circuit substrate arranged on an opposite side of a bottom surface of the recess portion with the Peltier element sandwiched therebetween, and a sensor element attached to an opposite side of a surface of the circuit substrate, the surface being opposed to the Peltier element. According to this, the circuit substrate lies between the Peltier element and the sensor element, and an inner side surface of the recess portion is present beside the Peltier element, thus suppressing the movement of foreign matters from the Peltier element to the sensor element.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is a plan view illustrating a configuration example of a sensor apparatus according to embodiment 1 of the present disclosure.

(2) FIG. **2** is a cross-sectional view illustrating the configuration example of the sensor apparatus according to embodiment 1 of the present disclosure.

(3) FIG. **3** is an exploded cross-sectional view illustrating the configuration example of the sensor apparatus according to embodiment 1 of the present disclosure.

(4) FIG. **4** is a plan view illustrating a configuration example of an upper surface side of a package substrate according to embodiment 1 of the present disclosure.

(5) FIG. **5** is a plan view illustrating a positional relation between the package substrate and a ceramic interposer substrate according to embodiment 1 of the present disclosure.

(6) FIG. **6** is a cross-sectional view illustrating a configuration example of a Peltier element according to embodiment 1 of the present disclosure.

(7) FIG. **7**A is a cross-sectional view illustrating a manufacturing method of the sensor apparatus according to embodiment 1 of the present disclosure.

(8) FIG. **7**B is a cross-sectional view illustrating the manufacturing method of the sensor apparatus according to embodiment 1 of the present disclosure.

(9) FIG. **7**C is a cross-sectional view illustrating the manufacturing method of the sensor apparatus according to embodiment 1 of the present disclosure.

(10) FIG. **7**D is a cross-sectional view illustrating the manufacturing method of the sensor apparatus according to embodiment 1 of the present disclosure.

(11) FIG. **7**E is a cross-sectional view illustrating the manufacturing method of the sensor apparatus according to embodiment 1 of the present disclosure.

(12) FIG. **8** is a cross-sectional view illustrating a configuration example of a sensor apparatus according to embodiment 2 of the present disclosure.

(13) FIG. **9** is a cross-sectional view illustrating a configuration example of a sensor apparatus according to embodiment 3 of the present disclosure.

(14) FIG. **10** is a cross-sectional view illustrating a configuration example of a sensor apparatus according to embodiment 4 of the present disclosure.

(15) FIG. **11** is a cross-sectional view illustrating a configuration example of a sensor apparatus according to embodiment 5 of the present disclosure.

(16) FIG. **12** is a plan view illustrating a configuration example of a sensor element according to embodiment 6 of the present disclosure.

(17) FIG. **13** is a cross-sectional view illustrating the configuration example of the sensor element according to embodiment 6 of the present disclosure.

(18) FIG. **14** is a circuit diagram illustrating a pixel circuit of each pixel of a sensor element

according to embodiment 7 of the present disclosure.
(19) FIG. **15** is a cross-sectional view illustrating a pixel structure of the sensor element according to embodiment 7 of the present disclosure.
DESCRIPTION OF EMBODIMENTS
(20) A description will be given below of embodiments of the present disclosure with reference to drawings. In the representation of the drawings referred to in the following description, the same or similar portions will be denoted by the same or similar reference signs. It should be noted, however, that the drawings are schematic and that a relation between thicknesses and planar dimensions, a ratio between the thicknesses of respective layers, and the like are different from the real ones. Accordingly, the following description should be considered to determine the specific thicknesses and dimensions. Also, it is a matter of course that the drawings include the portions where the relation between the dimensions and the ratio differ from one drawing to another.
(21) Also, a definition of directions such as upward and downward directions in the following description is merely for convenience of description and does not restrict a technical idea of the present disclosure. For example, it is a matter of course that if one observes a target by rotating it by 90 degrees, upward and downward directions are changed to leftward and rightward directions for interpretation and that if one observes the target by rotating it by 180 degrees, upward and downward directions are reversed for interpretation.
(22) Also, in the following description, there are cases where directions are described by using wordings “X-axis direction,” “Y-axis direction,” and “Z-axis direction.” For example, the Z-axis direction is the direction along the thickness of a sensor apparatus **100** which will be described later. The X-axis direction and the Y-axis direction are the directions orthogonal to the Z-axis direction. The X-axis direction, the Y-axis direction, and the Z-axis direction are orthogonal to one another. Also, in the following description, the term “plan view” refers to viewing from the Z-axis direction.
Embodiment 1
(23) FIG. **1** is a plan view illustrating a configuration example of the sensor apparatus **100** according to embodiment 1 of the present disclosure. FIG. **2** is a cross-sectional view illustrating the configuration example of the sensor apparatus **100** according to embodiment 1 of the present disclosure. FIG. **2** illustrates a cross-section obtained by cutting FIG. **1** along an X-Z plane that passes through an II-II′ line.
(24) As illustrated in FIGS. **1** and **2**, the sensor apparatus **100** includes a sensor assembly package **50** and a seal-glass-equipped lid **60** (example of a lid) attached to the side of an upper surface **50***a* of the sensor assembly package **50**. The sensor assembly package **50** includes a package substrate **10**, a Peltier element **20**, a ceramic interposer substrate **30** (example of a circuit substrate), and a sensor element **40**. A description will be given first of a configuration of the sensor assembly package **50**.
(25) FIG. **3** is an exploded cross-sectional view illustrating the configuration example of the sensor apparatus **100** according to embodiment 1 of the present disclosure. The package substrate **10** is a multi-layer substrate that includes ceramic such as alumina (aluminum oxide) and is, for example, a PGA (Pin Grid Array) substrate. As illustrated in FIG. **3**, the package substrate **10** has a first surface (e.g., upper surface **10***a*) and a second surface (e.g., lower surface **10***b*) on the opposite side of the first surface. In the package substrate **10**, plural wirings are provided therein in multiple layers between the upper surface **10***a* and the lower surface **10***b*. These wirings are connected to plural terminals (e.g., pin-shaped terminals **13**) provided on the lower surface **10***b* of the package substrate **10**.
(26) FIG. **4** is a plan view illustrating a configuration example of the side of the upper surface **10***a* of the package substrate **10** according to embodiment 1 of the present disclosure. As illustrated in FIGS. **3** and **4**, a cavity **11** is provided on the side of the upper surface **10***a* of the package substrate **10**. The cavity **11** has a first recess portion **111** and a second recess portion **112** provided on a

bottom surface **111***a* of the first recess portion **111** (examples of a recess portion). The plan-view shape of the first recess portion **111** and the second recess portion **112** is, for example, rectangular. The first recess portion **111** has an opening surface that is larger in diameter than that of the second recess portion **112**.
(27) The Peltier element **20** is arranged in the second recess portion **112** and is attached to a bottom surface **112***a* of the second recess portion **112**, for example, via an adhesive **51** (refer to FIG. **2**). An upper surface of the Peltier element **20** arranged in the second recess portion **112** (e.g., upper surface **25***a* of a second ceramic substrate **25** which will be described later) is at the same height or approximately at the same height as that of the bottom surface **111***a* of the first recess portion **111**.
(28) Pin-shaped terminals **12** for connection to lead wires of the Peltier element **20** are provided on the bottom surface **112***a* of the second recess portion **112**. The two pin-shaped terminals **12** are provided. One of the two pin-shaped terminals **12** is connected to a positive lead wire of the Peltier element **20**, and the other is connected to a negative lead wire of the Peltier element **20**.
(29) As illustrated in FIGS. **2** to **4**, a seal ring **15** is provided on the side of the upper surface **10***a* of a perimeter portion of the package substrate **10**. The seal ring **15** is continuously provided in such a manner as to surround the cavity **11** of the package substrate **10** in plan view. The seal ring **15** is a part that is joined to a metallic portion **63**, which will be described later, of the seal-glass-equipped lid **60**. The seal ring **15** is, for example, iron (Fe)-nickel (Ni)-cobalt (Co) alloy (what is called Kovar) and is surface-treated with Ni and gold (Au) plating or the like.
(30) FIG. **5** is a plan view illustrating a positional relation between the package substrate **10** and the ceramic interposer substrate **30** according to embodiment 1 of the present disclosure. As illustrated in FIG. **2**, the side of a lower surface **30***b* of the ceramic interposer substrate **30** is attached to the bottom surface **111***a* of the first recess portion **111** and the Peltier element **20** via the adhesive **51**. As illustrated in FIGS. **2** and **5**, the ceramic interposer substrate **30** is arranged in such a manner as to entirely cover an opening surface of the second recess portion **112**.
(31) Plural bonding pads **14** are provided in a region exposed from under the ceramic interposer substrate **30**, the region being the bottom surface **111***a* of the first recess portion **111**. Also, plural bonding pads **31** are provided on the side of the lower surface **30***b* of the ceramic interposer substrate **30**. At least some of the plural bonding pads **31** are connected to the bonding pads **14** via wires **54**. Also, at least some of the plural bonding pads **31** are connected to bonding pads **41** of the sensor element **40** via wires **55**. Alternatively, both the wires **54** and **55** may be connected to the single bonding pad **31**. The wires **54** and **55** are, for example, gold wires.
(32) The sensor element **40** is, for example, a CMOS (Complementary Metal Oxide Semiconductor) image sensor or a CCD (Charge Coupled Device) image sensor. A photoelectric conversion layer of the CMOS image sensor or the CCD (Charge Coupled Device) image sensor includes, for example, silicon (Si). Light detected by the sensor element **40** is not limited to visible light and may be, for example, infrared rays or ultraviolet rays. The bonding pads **41** are provided in a perimeter region on the side of the upper surface **40***a* of the sensor element **40**. The side of a lower surface **40***b* of the sensor element **40** is attached to the side of the lower surface **30***b* of the ceramic interposer substrate **30** via the adhesive **51**.
(33) FIG. **6** is a cross-sectional view illustrating a configuration example of the Peltier element **20** according to embodiment 1 of the present disclosure. As illustrated in FIG. **6**, the Peltier element **20** has a first ceramic substrate **21**, a first copper electrode **22**, a second ceramic substrate **25**, a second copper electrode **26**, a P-type thermoelectric semiconductor **27**, and an N-type thermoelectric semiconductor **28**. The first copper electrode **22** is provided on the first ceramic substrate **21**. The second ceramic substrate **25** is opposed to the first ceramic substrate **21**. The second copper electrode **26** is provided on the second ceramic substrate **25**. The P-type thermoelectric semiconductor **27** and the N-type thermoelectric semiconductor **28** are each arranged between the first ceramic substrate **21** and the second ceramic substrate **25**. The P-type thermoelectric semiconductor **27** and the N-type thermoelectric semiconductor **28** each have its one end connected

to the first copper electrode **22** and its other end connected to the second copper electrode **26**. The P-type thermoelectric semiconductor **27** and the N-type thermoelectric semiconductor **28** are connected alternately and in series via the first copper electrode **22** and the second copper electrode **26**.

(34) As illustrated in FIG. **6**, in the Peltier element **20**, when a direct current is passed from the N-type thermoelectric semiconductor **28**, the second ceramic substrate **25** absorbs heat T**1** (heat absorption), and the first ceramic substrate **21** radiates heat T**2** (heat radiation). The second ceramic substrate **25** is attached to the ceramic interposer substrate **30** via the adhesive **51**, and the first ceramic substrate **21** is attached to the package substrate **10** via the adhesive **51**. This makes it possible for the Peltier element **20** to dissipate heat generated by the sensor element **40** or the like from the ceramic interposer substrate **30** to the package substrate **10**.

(35) A description will be given next of a configuration of the sensor assembly package **50**. As illustrated in FIGS. **1** to **3**, the seal-glass-equipped lid **60** has a seal glass **61**, a ceramic frame **62**, and the metallic portion **63**. The ceramic frame **62** is provided on the side of a lower surface **61***b* of a perimeter portion of the seal glass **61**. The metallic portion **63** is provided on the side of a lower surface **62***b* of the ceramic frame **62**. The seal glass **61** and the ceramic frame **62** are joined together, for example, with low melting point glass. The ceramic frame **62** and the metallic portion **63** are joined together, for example, with an Ag—Cu wax material or the like.

(36) The metallic portion **63** is a part joined to the seal ring **15** of the package substrate **10**, for example, by seam welding or other means. The metallic portion **63** includes the same material as that of the seal ring **15**, one example of which is iron (Fe)-nickel (Ni)-cobalt (Co) alloy (what is called Kovar) and is surface-treated with Ni and gold (Au) plating or the like. The seal-glass-equipped lid **60** is joined to the side of the upper surface **50***a* of the sensor assembly package **50**, and the side of the upper surface **50***a* of the sensor assembly package **50** is sealed airtight.

(37) A description will be given next of a manufacturing method of the sensor apparatus **100**. It should be noted that various apparatuses such as an application apparatus for applying the adhesive **51**, an attachment apparatus for attaching the Peltier element **20**, the ceramic interposer substrate **30**, or the seal-glass-equipped lid **60**, a wire bonding apparatus, and a seam welding apparatus are used to manufacture the sensor apparatus **100**. In the embodiments of the present disclosure, these apparatuses will be collectively referred to as manufacturing apparatuses. Also, at least some of tasks performed by the manufacturing apparatuses may be performed by workers.

(38) FIGS. **7**A, **7**B, **7**C, **7**D, and **7**E are cross-sectional views illustrating the manufacturing method of the sensor apparatus **100** according to embodiment 1 of the present disclosure. First, as illustrated in FIG. **7**A, the manufacturing apparatus applies the adhesive **51** to the bottom surface **112***a* of the second recess portion **112** of the package substrate **10**. Next, as illustrated in FIG. **7**B, the manufacturing apparatus attaches the Peltier element **20** to the bottom surface **112***a* of the second recess portion **112** via the adhesive **51**. Also, before or after this bonding, the manufacturing apparatus wraps the lead wires of the Peltier element **20** around the pin-shaped terminals **12** and solders the lead wires. This allows the Peltier element **20** to be electrically connected to the package substrate **10**.

(39) Next, as illustrated in FIG. **7**C, the manufacturing apparatus attaches the ceramic interposer substrate **30** to the Peltier element **20** and the package substrate **10**. For example, the manufacturing apparatus applies the adhesive **51** to the bottom surface **111***a* of the first recess portion **111** and to the upper surface **25***a* of the second ceramic substrate **25** which is at the same height as that of the bottom surface **111***a*. Then, the manufacturing apparatus attaches the lower surface **30***b* of the ceramic interposer substrate **30** to the bottom surface **111***a* of the first recess portion **111** and to the upper surface **25***a* of the second ceramic substrate **25** via the adhesive **51**. This allows the opening surface of the second recess portion **112** to be entirely covered with the ceramic interposer substrate **30**. A path that runs through a space between the Peltier element **20** and the sensor element **40** is closed by the ceramic interposer substrate **30** and an inner side surface **112***c* of the second recess

portion **112**, thereby separating the space around the Peltier element **20** from the space around the sensor element **40**.

(40) Next, as illustrated in FIG. **7**D, the manufacturing apparatus attaches the sensor element **40** to an upper surface **30***a* of the ceramic interposer substrate **30** via the adhesive **51**. Then, the manufacturing apparatus electrically connects the bonding pads **41** of the sensor element **40** and the bonding pads **31** of the ceramic interposer substrate **30** via the wires **55**. Also, the manufacturing apparatus electrically connects the bonding pads **31** of the ceramic interposer substrate **30** and the bonding pads **14** of the package substrate **10** via the wires **55**. This completes the sensor assembly package **50**.

(41) Next, as illustrated in FIG. **7**E, with the seal-glass-equipped lid **60** and the sensor assembly package **50** aligned with each other, the manufacturing apparatus attaches the seal-glass-equipped lid **60** to the sensor assembly package **50**, for example, by seam welding or other means. This allows the space between the seal-glass-equipped lid **60** and the sensor assembly package (hereinafter referred to as a “package interior”) to be sealed airtight.

(42) Seam welding is a type of resistance welding and is a continuous welding method that uses a roller electrode and achieves welding by rotating the electrode while applying a pressure and passing a current. The seal ring **15** of the sensor assembly package **50** and the metallic portion **63** of the seal-glass-equipped lid **60** are joined by this method. By placing the inside of a chamber provided in the seam welding apparatus into a dry air, nitrogen, vacuum, or other atmosphere, this process can also keep the atmosphere inside the package in a dry air, nitrogen, or vacuum atmosphere.

(43) As described above, the sensor apparatus **100** according to embodiment 1 of the present disclosure includes the package substrate **10**, the Peltier element **20**, the ceramic interposer substrate **30**, and the sensor element **40**. The package substrate **10** has the second recess portion **112** on the side of the upper surface **10***a* and has the plural pin-shaped terminals **13** on the side of the lower surface **10***b* located on the opposite side of the upper surface **10***a*. The Peltier element **20** is arranged in the second recess portion **112**. The ceramic interposer substrate **30** is arranged on the opposite side of the bottom surface **112***a* of the second recess portion **112** with the Peltier element **20** sandwiched therebetween. In the ceramic interposer substrate **30**, the sensor element **40** is attached to the opposite side of the surface opposed to the Peltier element **20**.

(44) According to this, the ceramic interposer substrate **30** lies between the Peltier element **20** and the sensor element **40**, and the inner side surface **112***c* of the second recess portion **112** is present beside the Peltier element **20**. Also, there is no gap S between the inner side surface **112***c* of the second recess portion **112** and the ceramic interposer substrate **30**. The space around the Peltier element **20** is separated from the space around the sensor element **40** by the inner side surface **112***c* of the second recess portion **112** and the ceramic interposer substrate **30**, thus preventing the movement of foreign matters from the Peltier element **20** to the sensor element **40**. This makes it possible, even in a case where foreign matters such as dust adhere inside the Peltier element **20** or to its outer surface, to reduce the number of foreign matters that adhere to the sensor element **40** and to decrease the possibility that the operation of the sensor element **40** is affected due to foreign matters.

(45) Also, the sensor apparatus **100** further includes the seal-glass-equipped lid **60** that is attached to the package substrate **10** and covers the ceramic interposer substrate **30** and the sensor element **40**. The space between the seal-glass-equipped lid **60** and the package substrate **10** is sealed airtight. According to this, it is possible to prevent intrusion of foreign matters from outside into the sensor apparatus **100**. It is possible to further reduce the number of foreign matters that adhere to the sensor element **40**.

(46) Also, the seal-glass-equipped lid **60** is attached to the package substrate **10** by seam welding. According to this, it is possible to improve reliability of airtight seal inside the package.

(47) Also, the package substrate **10** is a multi-layer substrate that includes ceramic. The seal-glass-

equipped lid **60** includes the seal glass **61**, the ceramic frame **62**, and the metallic portion **63**. Neither of the package substrate **10** and the seal-glass-equipped lid **60** included in the package of the sensor apparatus **100** includes resin. This makes it possible to prevent the intrusion of moisture and the like from outside into the sensor apparatus **100** with high reliability.

Embodiment 2

(48) In above embodiment 1, it has been described that the ceramic interposer substrate **30** is arranged in such a manner as to entirely cover the opening surface of the second recess portion **112** and that the space around the Peltier element **20** is separated from the space around the sensor element **40** by the ceramic interposer substrate **30** and the inner side surface **112***c* of the second recess portion **112**. However, the embodiments of the present disclosure are not limited thereto. For example, the ceramic interposer substrate **30** may be arranged in such a manner as to partially cover the opening surface of the second recess portion **112**. The Peltier element **20** and the sensor element **40** may not be completely spatially separated.

(49) FIG. **8** is a cross-sectional view illustrating a configuration example of a sensor apparatus **100**A according to embodiment 2 of the present disclosure. As illustrated in FIG. **8**, in the sensor apparatus **100**A, the ceramic interposer substrate **30** partially, and not completely, covers the opening surface of the second recess portion **112**. A plan-view size of the ceramic interposer substrate **30** is approximately equal to or slightly smaller than that of the opening surface of the second recess portion **112**. Because of this structure, the gap S is present between the inner side surface **112***c* of the second recess portion **112** and an edge portion **30***e* of the ceramic interposer substrate **30**. There is a possibility that foreign matters may pass through the gap S.

(50) However, the ceramic interposer substrate **30** protrudes more outward than the Peltier element **20**, and the path that runs through the space between the Peltier element **20** and the sensor element **40** detours the edge portion **30***e* of the ceramic interposer substrate **30**. This lengthens a distance of the path that runs through the above space. Also, the inner side surface **112***c* of the second recess portion **112** is present beside the Peltier element **20**. The inner side surface **112***c* of the second recess portion **112** is in proximity to the edge portion **30***e* of the ceramic interposer substrate **30**, and the gap S is narrower.

(51) As described above, the path that runs through the above space has a long distance, and further, the gap S is narrower, thus suppressing the movement of foreign matters from the Peltier element **20** to the sensor element **40**. This makes it possible, even in the case where foreign matters such as dust adhere inside the Peltier element **20** or to its outer surface, to reduce the number of foreign matters that adhere to the sensor element **40** and to decrease the possibility that the operation of the sensor element **40** is affected due to foreign matters.

(52) Also, it is preferred that, in the sensor apparatus **100**A, the upper surface **30***a* of the ceramic interposer substrate **30** should be at the same or approximately at the same height as that of the bottom surface **111***a* of the first recess portion **111**, as illustrated in FIG. **8**. This can not only reduce the above gap S but also eliminate a difference in level between the bonding pads **14** and **31**, thus making it possible to shorten the length of the wires **54** that electrically connect the bonding pads **14** and **31**. As a result, it becomes possible to reduce an amount of Au used that is included in the wires **54**, thus contributing to reduced manufacturing cost of the sensor apparatus **100**A.

Embodiment 3

(53) In above embodiment 1, it has been described that the space around the Peltier element **20** is separated from the space around the sensor element **40** by the ceramic interposer substrate **30** and the inner side surface **112***c* of the second recess portion **112**. In the embodiments of the present disclosure, however, the structure for separating the space between the Peltier element **20** and the sensor element **40** is not limited to that described above.

(54) FIG. **9** is a cross-sectional view illustrating a configuration example of a sensor apparatus **100**B according to embodiment 3 of the present disclosure. As illustrated in FIG. **9**, the sensor apparatus **100**B includes a metallic casing **110** that is arranged in the second recess portion **112** and

that accommodates the Peltier element **20** therein. The casing **110** is, for example, a cuboid and shaped in such a manner that the side of a surface opposed to the bottom surface **112***a* of the second recess portion **112** is open. The inside of the casing **110** is fastened to the Peltier element **20** via the adhesive **51**, and the outside of the casing **110** is fastened to the ceramic interposer substrate **30** via the adhesive **51**. The space around the Peltier element **20** is separated from the space around the sensor element **40** by the casing **110**.
(55) In such a configuration, the casing **110** prevents the movement of foreign matters from the Peltier element **20** to the sensor element **40**. This makes it possible, even in the case where foreign matters such as dust adhere inside the Peltier element **20** or to its outer surface, to reduce the number of foreign matters that adhere to the sensor element **40** and to decrease the possibility that the operation of the sensor element **40** is affected due to foreign matters.
(56) Also, the casing **110** is metallic and covers the Peltier element **20**. This allows the casing **110** to shield the Peltier element **20**, thus suppressing electromagnetic waves (noise) generated by the Peltier element **20** from entering the sensor element **40**.
(57) A description will be given of a process for attaching the casing **110**. For example, after the Peltier element **20** is attached to the bottom surface **112***a* of the second recess portion **112** and the lead wires of the Peltier element **20** are electrically connected to the pin-shaped terminals **12**, the manufacturing apparatus applies the adhesive **51** to the upper surface of the Peltier element **20** and places the casing **110** over the Peltier element **20** to which the adhesive **51** has been applied. This allows the casing **110** to be fastened to the upper surface of the Peltier element **20** via the adhesive **51**. Thereafter, the manufacturing apparatus applies the adhesive **51** to an upper surface of the casing **110** and attaches the ceramic interposer substrate **30** to the upper surface of the casing **110** to which the adhesive **51** has been applied. The subsequent processes are the same as in embodiment 1.
Embodiment 4
(58) In the embodiments of the present disclosure, the structure for separating the space between the Peltier element **20** and the sensor element **40** may be realized by a component other than the casing.
(59) FIG. **10** is a cross-sectional view illustrating a configuration example of a sensor apparatus **100**C according to embodiment 4 of the present disclosure. As illustrated in FIG. **10**, the sensor apparatus **100**C includes a wall member **120** that is sandwiched between the bottom surface **112***a* of the second recess portion **112** and the ceramic interposer substrate **30** and laterally surrounds the Peltier element **20**. For example, the wall member **120** includes resin. The plan-view shape of the wall member **120** is a frame, and the Peltier element **20** is arranged inside the frame. The space around the Peltier element **20** is separated from the space around the sensor element **40** by the ceramic interposer substrate **30** and the wall member **120**.
(60) In such a configuration, the ceramic interposer substrate **30** and the wall member **120** prevent the movement of foreign matters from the Peltier element **20** to the sensor element **40**. This makes it possible, even in the case where foreign matters such as dust adhere inside the Peltier element **20** or to its outer surface, to reduce the number of foreign matters that adhere to the sensor element **40** and to decrease the possibility that the operation of the sensor element **40** is affected due to foreign matters.
(61) It should be noted that, in the embodiments of the present disclosure, the material included in the wall member **120** is not limited to resin and may be a metal. In a case where the wall member **120** includes a metal, the wall member **120** can laterally shield the Peltier element **20**, thus reducing noise that propagates horizontally (e.g., direction parallel to the X-Y plane).
(62) A description will be given of a process for forming the wall member **120**. For example, after the Peltier element **20** is attached to the bottom surface **112***a* of the second recess portion **112** and the lead wires of the Peltier element **20** are electrically connected to the pin-shaped terminals **12**, the manufacturing apparatus forms the wall member **120** on the bottom surface **112***a* in such a

manner as to surround the periphery of the Peltier element **20**. Although the wall member **120** may be formed in any manner, an example of forming method is to attach the wall member **120**, which is formed in advance by resin molding, to the bottom surface **112***a* via the adhesive (not illustrated). Thereafter, the manufacturing apparatus attaches the ceramic interposer substrate **30** to the upper surface of the Peltier element **20** via the adhesive **51**. The adhesive **51** may or may not be present between the wall member **120** and the ceramic interposer substrate **30**. The subsequent processes are the same as in embodiment 1.

Embodiment 5

(63) In the embodiments of the present disclosure, the structure for reducing noise may be provided on the ceramic interposer substrate **30**.

(64) FIG. **11** is a cross-sectional view illustrating a configuration example of a sensor apparatus **100**D according to embodiment 5 of the present disclosure. As illustrated in FIG. **11**, in the sensor apparatus **100**D, a magnetic shield pattern **130** (example of a conductor layer) is provided on the side of the lower surface **30***b* of the ceramic interposer substrate **30**. The magnetic shield pattern **130** is provided on the side of the lower surface **30***b* of the ceramic interposer substrate **30** in such a manner as to surround a region opposed to the Peltier element **20** on the lower surface **30***b* of the ceramic interposer substrate **30**. The magnetic shield pattern **130** is fixed at a given potential (e.g., ground voltage: 0V). In such a configuration, the magnetic shield pattern **130** can shield the Peltier element **20**, thus suppressing noise generated by the Peltier element **20** from entering the sensor element **40**.

Embodiment 6

(65) In above embodiments 1 to 5, the sensor element **40** has been illustrated as an example of a “sensor element” of the present disclosure, and it has been described that the sensor element **40** is a CMOS image sensor or a CCD image sensor. However, the “sensor element” of the present disclosure is not limited to a CMOS image sensor or a CCD image sensor. Also, the material included in the photoelectric conversion layer of the “sensor element” of the present disclosure is not limited to silicon (Si) and may be a compound semiconductor.

(66) For example, the “sensor element” of the present disclosure may be an InGaAs sensor whose photoelectric conversion layer includes a group III-V semiconductor such as InGaAs (indium-gallium-arsenide) and may be, for example, a sensor element **200** illustrated in FIGS. **12** and **13**. Even in such a mode, it is possible to suppress the movement of foreign matters from the Peltier element **20** to the sensor element **200**. A description will be given below of a configuration example of the sensor element **200**.

(67) FIGS. **12** and **13** are a plan view and a cross-sectional view illustrating a configuration example of the sensor element **200** according to embodiment 6 of the present disclosure. FIG. **12** illustrates a planar configuration of the sensor element **200**, and FIG. **13** illustrates a cross-sectional configuration thereof along line B-B′ in FIG. **12**. The sensor element **200** is applied to, for example, an infrared sensor using a compound semiconductor material such as a group III-V semiconductor and has a function to photoelectrically convert light of a wavelength ranging from a visible region (e.g., 380 nm or more to less than 780 nm) to a short-infrared region (e.g., 780 nm or more to less than 2400 nm). In the sensor element **200**, plural light reception unit regions p (pixels P) that are, for example, arranged two-dimensionally are provided (FIG. **13**).

(68) The sensor element **200** has an element region R**1** in the center and a peripheral region R**2** that is provided in the outside of the element region R**1** and that surrounds the element region R**1** (FIG. **12**). The sensor element **200** has a conductive film **215**B that is provided to span from the element region R**1** to the peripheral region R**2**. The conductive film **215**B has an opening in a region opposed to a center portion of the element region R**1**.

(69) The sensor element **200** has a layered structure in which an element substrate **210** and a readout circuit substrate **220** are stacked (FIG. **13**). One surface of the element substrate **210** is a light entry surface (light entry surface S**1**), and the surface on the opposite side of the light entry

surface S**1** (other surface) is a junction surface with the readout circuit substrate **220** (junction surface S**2**).
(70) The element substrate **210** has a wiring layer **210**W, first electrodes **211**, a semiconductor layer **210**S (first semiconductor layer), a second electrode **215**, and a passivation film **216** in sequence from the position close to the readout circuit substrate **220**. A surface of the semiconductor layer **210**S, the surface being opposed to the wiring layer **210**W, and an edge surface (side surface) thereof are covered with an insulating film **217**. The readout circuit substrate **220** is what is called a ROIC (Readout integrated circuit) and has a wiring layer **220**W, a multi-layer wiring layer **222**C, and a semiconductor substrate **221**. The wiring layer **220**W is in contact with the junction surface S**2** of the element substrate **210**. The semiconductor substrate **221** is opposed to the element substrate **210** with the wiring layer **220**W and the multi-layer wiring layer **222**C therebetween.
(71) The element substrate **210** has the semiconductor layer **210**S in the element region R**1**. In other words, a region where the semiconductor layer **210**S is provided is the element region R**1** of the sensor element **200**. Of the element region R**1**, a region exposed from the conductive film **215**B (region opposed to the opening of the conductive film **215**B) is a light reception region. Of the element region R**1**, a region covered with the conductive film **215**B is an OPB (Optical Black) region R**1**B. The OPB region R**1**B is provided in such a manner as to surround the light reception region. The OPB region R**1**B is used to acquire a black level pixel signal.
(72) The element substrate **210** has a filling-in layer **218** together with the insulating film **217** in the peripheral region R**2**. Holes H**1** and H**2** that penetrate the element substrate **210** and reach the readout circuit substrate **220** are provided in the peripheral region R**2**. In the sensor element **200**, light enters the semiconductor layer **210**S from the light entry surface S**1** of the element substrate **210** via the passivation film **216**, the second electrode **215**, and a second contact layer **214**. A signal charge acquired as a result of photoelectric conversion by the semiconductor layer **210**S moves via the first electrodes **211** and the wiring layer **210**W and is read out by the readout circuit substrate **220**. A description will be given below of the configuration of each portion.
(73) The wiring layer **210**W is provided to span from the element region R**1** to the peripheral region R**2** and has the junction surface S**2** with the readout circuit substrate **220**. In the sensor element **200**, the junction surface S**2** of the element substrate **210** is provided in the element region R**1** and the peripheral region R**2**, and, for example, the junction surface S**2** of the element region R**1** and the junction surface S**2** of the peripheral region R**2** form the same plane. In the sensor element **200**, the junction surface S**2** of the peripheral region R**2** is formed as a result of the provision of the filling-in layer **218**.
(74) The wiring layer **210**W has contact electrodes **219**E and dummy electrodes **219**ED, for example, inside interlayer insulating films **219**A and **219**B. For example, the interlayer insulating film **219**B is provided on the side of the readout circuit substrate **220**, and an interlayer insulating film **19**A is provided on the side of a first contact layer **212**, and the interlayer insulating films **219**A and **219**B are stacked one on top of another. The interlayer insulating films **219**A and **219**B each include, for example, an inorganic insulating material. Silicon nitride (SiN), aluminum oxide (Al_2O_3), silicon oxide (SiO_2), hafnium oxide (HfO_2), and the like are examples of this inorganic insulating material. The interlayer insulating films **219**A and **219**B may include the same inorganic insulating material.
(75) The contact electrodes **219**E are provided, for example, in the element region R**1**. The contact electrodes **219**E are used to electrically connect the first electrodes **211** and the readout circuit substrate **220** and are each provided for each pixel P in the element region R**1**. The contact electrodes **219**E adjacent to each other are electrically separated from each other by the filling-in layer **218** and the interlayer insulating films **219**A and **219**B. The contact electrodes **219**E each include, for example, a copper (Cu) pad and are exposed on the junction surface S**2**. The dummy electrodes **219**ED are provided, for example, in the peripheral region R**2**. The dummy electrodes **219**ED are connected to dummy electrodes **222**ED of the wiring layer **220**W. By providing the

dummy electrodes **219**ED and the dummy electrodes **222**ED, it is possible to improve strength of the peripheral region R**2**. The dummy electrodes **219**ED are formed, for example, by the same process as that of the contact electrodes **219**E. The dummy electrodes **219**ED each include, for example, a copper (Cu) pad and are exposed on the junction surface S**2**.
(76) The first electrodes **211** provided between the contact electrodes **219**E and the semiconductor layer **210**S are electrodes (anodes) to which a voltage for reading out a signal charge (holes or electrons; a description will be given hereinafter assuming that the signal charge is holes for reasons of convenience) generated by a photoelectric conversion layer **213** and are each provided for each pixel P in the element region R**1**. The first electrodes **211** are provided in such a manner as to fill in the openings of the insulating film **217** and are in contact with the semiconductor layer **210**S (more specifically, a diffusion region **212**A). The first electrodes **211** are, for example, larger than the openings of the insulating film **217**, and part of the first electrodes **211** is provided in the filling-in layer **218**. That is, an upper surface of the first electrodes **211** (the surface on the side of the semiconductor layer **210**S) is in contact with the diffusion region **212**A, and part of a lower surface and a side surface of the first electrodes **211** is in contact with the filling-in layer **218**. The first electrodes **211** adjacent to each other are electrically separated from each other by the insulating film **217** and the filling-in layer **218**.
(77) The first electrodes **211** include, for example, any one of titanium (Ti), tungsten (W), titanium nitride (TiN), platinum (Pt), gold (Au), germanium (Ge), palladium (Pd), zinc (Zn), nickel (Ni), and aluminum (Al) or an alloy that includes at least one of them. The first electrodes **211** may be a single film of one of these materials or a layered film that includes a combination of two or more of these materials. For example, the first electrodes **211** include a layered film of titanium and tungsten. The first electrodes **211** are, for example, several tens to hundreds of nanometers in thickness.
(78) The semiconductor layer **210**S includes, for example, the first contact layer **212**, the photoelectric conversion layer **213**, and the second contact layer **214** from the position close to the wiring layer **210**W. The first contact layer **212**, the photoelectric conversion layer **213**, and the second contact layer **214** have the same planar shape, and edge surfaces of the respective layers are arranged at the same position in plan view.
(79) The first contact layer **212** is provided, for example, in common for all the pixels P and arranged between the insulating film **217** and the photoelectric conversion layer **213**. The first contact layer **212** is used to electrically separate the pixels P adjacent to each other, and the plural diffusion regions **212**A, for example, are provided in the first contact layer **212**. By using, to form the first contact layer **212**, a compound semiconductor material having a bandgap larger than that of the compound semiconductor material included in the photoelectric conversion layer **213**, it is also possible to suppress dark current. For example, N-type InP (indium phosphide) can be used for the first contact layer **212**.
(80) The diffusion regions **212**A provided in the first contact layer **212** are provided apart from each other. The diffusion regions **212**A are each provided for each pixel P, and the first electrode **211** is connected to each of the diffusion regions **212**A. The diffusion region **212**A is also provided in the OPB region R**1**B. The diffusion regions **212**A are used to read out the signal charge generated by the photoelectric conversion layer **213** for each pixel P and each include, for example, a P-type impurity. Zn (zinc) or the like is an example of a P-type impurity. As described above, a pn junction interface is formed between the diffusion region **212**A and the first contact layer **212** other than the diffusion region **212**A, to electrically separate the pixels P adjacent to each other. The diffusion regions **212**A are provided, for example, in the direction of the thickness of the first contact layer **212** and also provided in part of the direction of the thickness of the photoelectric conversion layer **213**.
(81) The photoelectric conversion layer **213** between the first electrodes **211** and the second electrode **215**, and more specifically, the photoelectric conversion layer **213** between the first

contact layer **212** and the second contact layer **214**, is provided, for example, in common for all the pixels P. The photoelectric conversion layer **213** generates a signal charge by absorbing light having a specific wavelength and includes, for example, a compound semiconductor material such as an i-type group III-V semiconductor. InGaAs (indium-gallium-arsenic), InAsSb (indium-arsenic-antimony), InAs (indium-arsenic), InSb (indium-antimony), HgCdTe (mercury-cadmium-tellurium), and the like are examples of a compound semiconductor material included in the photoelectric conversion layer **213**. The photoelectric conversion layer **213** may include Ge (germanium). The photoelectric conversion layer **213** photoelectrically converts, for example, light of a wavelength ranging from the visible region to the short-infrared region.

(82) The second contact layer **214** is provided, for example, in common for all the pixels P. The second contact layer **214** is provided between the photoelectric conversion layer **213** and the second electrode **215** and is in contact with them. The second contact layer **214** is a region where the charge emitted by the second electrode **215** moves and includes, for example, a compound semiconductor including an N-type impurity. N-type InP (indium phosphide), for example, can be used for the second contact layer **214**.

(83) The second electrode **215** is provided on the second contact layer **214** (light entry side) in such a manner as to be in contact with the second contact layer **214**, for example, as a common electrode for each pixel P. The second electrode **215** is used to emit, of the charge generated by the photoelectric conversion layer **213**, the charge that is not used as a signal charge (cathode). For example, in a case where holes are read out from the first electrodes **211** as a signal charge, electrons, for example, can be emitted through the second electrode **215**. The second electrode **215** includes, for example, a conductive film that allows passage of incident light such as infrared rays. ITO (Indium Tin Oxide), ITiO (In.sub.2O.sub.3—TiO.sub.2), or the like can be used, for example, for the second electrode **215**. The second electrode **215** may be provided, for example, in a lattice pattern to partition the pixels P adjacent to each other. A conductive material having low light transmissivity can be used for the second electrode **215**.

(84) The passivation film **216** covers the second electrode **215** from the side of the light entry surface S**1**. The passivation film **216** may have an anti-reflection function. For example, silicon nitride (SiN), aluminum oxide (Al.sub.2O.sub.3), silicon oxide (SiO.sub.2), tantalum oxide (Ta.sub.2O.sub.3), and the like can be used for the passivation film **216**. The passivation film **216** has an opening **216**H in the OPB region R**1**B. The opening **216**H is provided in the shape of a picture frame surrounding the light reception region (FIG. **12**). The opening **216**H may be, for example, a rectangular or circular hole in plan view. The conductive film **215**B is electrically connected to the second electrode **215** by the opening **216**H of the passivation film **216**.

(85) The insulating film **217** is provided between the first contact layer **212** and the filling-in layer **218**, covers the edge surface of the first contact layer **212**, the edge surface of the photoelectric conversion layer **213**, the edge surface of the second contact layer **214**, and the edge surface of the second electrode **215**, and is in contact with the passivation film **216** in the peripheral region R**2**. The insulating film **217** includes oxide such as silicon oxide (SiO.sub.x) or aluminum oxide (Al.sub.2O.sub.3). The insulating film **217** may have a layered structure that includes plural films. The insulating film **217** may include a silicon-based insulating film such as silicon oxide nitride (SiON), silicon oxide including carbon (SiOC), silicon nitride (SiN), and silicon carbide (SiC). The insulating film **217** is, for example, several tens to hundreds of nanometers in thickness.

(86) The conductive film **215**B is provided to span from the OPB region R**1**B to a hole H**1** in the peripheral region R**2**. The conductive film **215**B is in contact with the second electrode **215** in the opening **216**H of the passivation film **216** provided in the OPB region R**1**B and is in contact with a wiring **222**CB of the readout circuit substrate **220** via the hole H**1**. This allows a voltage to be supplied to the second electrode **215** from the readout circuit substrate **220** via the conductive film **215**B. The conductive film **215**B not only functions as a voltage supply path to the second electrode **215** as described above but also has a function as a light-shielding film and forms the

OPB region R**1**B. The conductive film **215**B includes, for example, a metallic material that includes tungsten (W) aluminum (Al), titanium (Ti), molybdenum (Mo), tantalum (Ta), or copper (Cu). A passivation film may be provided on the conductive film **215**B.
(87) An adhesive layer B may be provided between the edge portion of the second contact layer **214** and the second electrode **215**. The adhesive layer B is used to form the sensor element **200** and plays a role of joining the semiconductor layer **210**S to a temporary substrate. The adhesive layer B includes, for example, tetraethoxysilane (TEOS), silicon oxide (SiO.sub.2), or the like. The adhesive layer B is provided, for example, wider than the edge surface of the semiconductor layer **210**S and is covered with the filling-in layer **218** together with the semiconductor layer **210**S. The insulating film **217** is provided between the adhesive layer B and the filling-in layer **218**.
Embodiment 7
(88) The "sensor element" of the present disclosure may be a sensor element **300** illustrated in FIGS. **14** and **15**. Even in such a mode, it is possible to suppress the movement of foreign matters from the Peltier element **20** (refer to FIG. **2**) to the sensor element **300**. A description will be given below of a configuration example of the sensor element **300**.
(89) FIG. **14** is a circuit diagram illustrating a pixel circuit of each pixel **302** of the sensor element **300** according to embodiment 7 of the present disclosure. FIG. **15** is a cross-sectional view illustrating a pixel structure of the sensor element **300** according to embodiment 7 of the present disclosure. A description will be given first of the pixel circuit illustrated in FIG. **14**.
(90) As illustrated in FIG. **14**, each pixel **302** includes a photoelectric conversion section **321**, a capacitive element **322**, a reset transistor **323**, an amplifying transistor **324**, and a selection transistor **325**. The photoelectric conversion section **321** includes a semiconductor thin film using a compound semiconductor such as InGaAs and generates a charge (signal charge) proportional to an amount of light received. A given bias voltage Va is applied to the photoelectric conversion section **321**.
(91) The capacitive element **322** stores the charge generated by the photoelectric conversion section **321**. The capacitive element **322** can include, for example, at least one of P-N junction capacitance, MOS capacitance, or wiring capacitance. When the reset transistor **323** is turned ON by a reset signal RST, the charge stored in the capacitive element **322** is emitted to a source (ground), thus resetting a potential of the capacitive element **322**.
(92) The amplifying transistor **324** outputs a pixel signal proportional to the stored potential of the capacitive element **322**. That is, the amplifying transistor **324** forms a source follower circuit with a load MOS (not illustrated) as a constant current source connected via a vertical signal line **309**. The pixel signal indicating a level proportional to the charge stored in the capacitive element **322** is output from the amplifying transistor **324** to a signal processing circuit (not illustrated) via the selection transistor **325**.
(93) A description will be given next of the pixel structure illustrated in FIG. **15**. It should be noted that each pixel **302** in a pixel array region can be classified into a normal pixel **302**A and a charge ejection pixel **302**B, depending on the difference in the manner of controlling the reset transistor **323**. It should be noted, however, that the normal pixel **302**A and the charge ejection pixel **302**B have the same pixel structure. Accordingly, these pixels will be simply described as the pixels **302**. It should be noted that the charge ejection pixel **302**B is arranged at an outermost side of the pixel array region.
(94) The readout circuit for the photoelectric conversion section **321**, the capacitive element **322**, the reset transistor **323**, the amplifying transistor **324**, and the selection transistor **325** of each pixel **302** illustrated in FIG. **14** is formed for each pixel in a semiconductor substrate **312** that includes a monocrystalline material such as monocrystalline silicon (Si). It should be noted that reference signs such as the capacitive element **322**, the reset transistor **323**, the amplifying transistor **324**, and the selection transistor **325** illustrated in FIG. **14** are omitted in FIG. **15**.
(95) As illustrated in FIG. **15**, an N-type semiconductor thin film **341** that will be the photoelectric

conversion section **321** (refer to FIG. **14**) is formed over the entire surface of the pixel array region on the upper side of the semiconductor substrate **312** which is the light entry side. InGaP, InAIP, InGaAs, and InAlAs may be used as the N-type semiconductor thin film **341**, and a compound semiconductor having a chalcopyrite structure may also be used. The compound semiconductor having a chalcopyrite structure is a material that offers a high light absorption coefficient and high sensitivity over a wide wavelength range and is preferably used as the N-type semiconductor thin film **341** for photoelectric conversion. Such a compound semiconductor having a chalcopyrite structure includes elements around group IV elements such as Cu, Al, GaIn, S, and Se, and CuGaInS-based mixed crystal, CuAlGaInS-based mixed crystal, CuAlGaInSSe-based mixed crystal, and the like are examples thereof.

(96) Also, in addition to the compound semiconductor described above, amorphous silicon (Si), germanium (Ge), a quantum-dot photoelectric conversion film, an organic photoelectric conversion film, and the like can also be used as the material of the N-type semiconductor thin film **341**. In the present embodiment, it is assumed that an InGaAs compound semiconductor is used as the N-type semiconductor thin film **341**.

(97) A high-concentration P-type layer **342** included in a pixel electrode is formed for each pixel on the lower side of the N-type semiconductor thin film **341**, the lower side being the side of the semiconductor substrate **312**. Then, an N-type layer **343** as a pixel separation region for separating the pixels **302** is formed between the high-concentration P-type layers **342** that are each formed for each pixel. The N-type layers **343** are each formed by using, for example, a compound semiconductor such as InP. The N-type layers **343** play a role of preventing dark current in addition to their function as the pixel separation region.

(98) Meanwhile, a high-concentration N-type layer **344** having higher concentration than the N-type semiconductor thin film **341** is formed on the upper side of the N-type semiconductor thin film **341**, the upper side being the light entry side. The high-concentration N-type layer **344** is formed by using the compound semiconductor such as InP that is used as the pixel separation region. The high-concentration N-type layer **344** functions as a barrier layer for preventing a charge generated by the N-type semiconductor thin film **341** from flowing in reverse. A compound semiconductor such as InGaAs, InP, and InAlAs can be used as the material of the high-concentration N-type layer **344**, for example.

(99) An anti-reflection film **45** is formed on the high-concentration N-type layer **344** being as the barrier layer. Silicon nitride (SiN), hafnium oxide (HfO.sub.2), aluminum oxide (Al.sub.2O.sub.3), zirconium oxide (ZrO.sub.2), tantalum oxide (Ta.sub.2O.sub.5), titanium oxide (TiO.sub.2), and the like can be used as the material of an anti-reflection film **345**, for example.

(100) Any one of the high-concentration N-type layer **344** or the anti-reflection film **345** also functions as an upper electrode of the two electrodes that vertically sandwich the N-type semiconductor thin film **341**, and the given voltage Va is applied to the high-concentration N-type layer **344** or the anti-reflection film **345** as the upper electrode.

(101) Color filters **346** and an on-chip lens **347** are further formed on the anti-reflection film **345**. The color filters **346** allow passage of any one of R (red), G (green), or B (blue) light (wavelength light) and are arranged, for example, in what is called a Bayer pattern in the pixel array region.

(102) A passivation layer **351** and an insulating layer **352** are formed on the lower side of the N-type layer **343** being as the pixel separation region and of the high-concentration N-type layer **344** included in the pixel electrode. Then, connection electrodes **353**A and **353**B and a bump electrode **354** are formed in such a manner as to penetrate the passivation layer **351** and the insulating layer **352**. The connection electrodes **353**A and **353**B and the bump electrode **354** electrically connect the high-concentration P-type layer **342** included in the pixel electrode and the capacitive element **322** that stores the charge.

Other Embodiments

(103) As described above, the present disclosure has been described by using the embodiments and

modification examples. However, the statement and the drawings which are part of this disclosure should not be construed as limiting the present disclosure. Various alternative embodiments, working examples, and operational technologies will become apparent to a person skilled in the art from this disclosure. In the embodiments of the present disclosure, for example, a heatsink may be attached to the package substrate **10** to improve heat radiation of the Peltier element **20** to the package substrate **10**. A region that is on the side of the lower surface **10***b* of the package substrate **10** and that is opposed to the Peltier element **20** is an example of a region where the heatsink is attached.
(104) As described above, it is a matter of course that the present technology includes various embodiments and the like not described herein. At least one of various omissions, substitutions, and alterations can be made to components without departing from the gist of the embodiments and modification examples described above. Also, advantageous effects described in the present specification are merely illustrative and not restrictive, and there may be additional advantageous effects.
(105) It should be noted that the present disclosure can also have the following configurations.
(106) (1)
(107) A sensor apparatus including: a package substrate having a recess portion on a side of a first surface and plural terminals on a side of a second surface located on an opposite side of the first surface; a Peltier element arranged in the recess portion; a circuit substrate arranged on an opposite side of a bottom surface of the recess portion with the Peltier element sandwiched therebetween; and a sensor element attached to an opposite side of a surface of the circuit substrate, the surface being opposed to the Peltier element.
(2)
(108) The sensor apparatus of feature (1), further including: a lid that is attached to the package substrate and covers the circuit substrate and the sensor element, in which a space between the lid and the package substrate is sealed airtight.
(3)
(109) The sensor apparatus of feature (2), in which the lid is attached to the package substrate by seam welding.
(4)
(110) The sensor apparatus of any one of features (1) to (3), in which a space around the Peltier element is separated from a space around the sensor element by the circuit substrate and an inner side surface of the recess portion.
(5)
(111) The sensor apparatus of any one of features (1) to (3), further including: a casing that is arranged in the recess portion and accommodates the Peltier element, in which a space around the Peltier element is separated from a space around the sensor element by the casing.
(6)
(112) The sensor apparatus of feature (5), in which the casing is metallic.
(7)
(113) The sensor apparatus of any one of features (1) to (3), further including: a wall member that is sandwiched between the bottom surface of the recess portion and the circuit substrate and laterally surrounds the Peltier element, in which a space around the Peltier element is separated from a space around the sensor element by the circuit substrate and the wall member.
(8)
(114) The sensor apparatus of any one of features (1) to (5) and (7), in which the circuit substrate has a conductor layer provided in such a manner as to surround a region opposed to the Peltier element.
(9)
(115) The sensor apparatus of any one of features (1) to (8), in which the sensor element has a

photoelectric conversion layer that includes silicon.
(10)
(116) The sensor apparatus of any one of features (1) to (8), in which the sensor element has a photoelectric conversion layer that includes a compound semiconductor.
REFERENCE SIGNS LIST
(117) **10**: Package substrate **10***a*, **25***a*, **30***a*, **40***a*: Upper surface **10***b*, **30***b*, **40***b*, **62***b*: Lower surface **11**: Cavity **12**, **13**: Pin-shaped terminal **14**: Bonding pad **15**: Seal ring **20**: Peltier element **21**: First ceramic substrate **22**: First copper electrode **25**: Second ceramic substrate **26**: Second copper electrode **27**: P-type thermoelectric semiconductor **28**: N-type thermoelectric semiconductor **30**: Ceramic interposer substrate **30***e*: Edge portion **31**: Bonding pad **40**, **200**, **300**: Sensor element **41**: Bonding pad **50**: Sensor assembly package **50***a*: Upper surface **51**: Adhesive **54**, **55**: Wire **60**: Seal-glass-equipped lid **61***b*: Lower surface **62**: Ceramic frame **63**: Metallic portion **100**, **100**A, **100**B, **100**C, **100**D: Sensor apparatus **110**: Casing **111**: First recess portion **111***a*, **112***a*: Bottom surface **112**: Second recess portion **112***c*: Inner side surface **120**: Wall member **130**: Magnetic shield pattern **210**: Element substrate **210**S: Semiconductor layer **210**W: Wiring layer **211**: First electrodes **212**: First contact layer **212**A: Diffusion region **213**: Photoelectric conversion layer **214**: Second contact layer **215**: Second electrode **215**B: Conductive film **216**: Passivation film **216**H: Opening **217**: Insulating film **218**: Filling-in layer **219**A, **219**B: Interlayer insulating film **219**E: Contact electrode **219**ED: Dummy electrode **220**: Readout circuit substrate **220**W: Wiring layer **221**: Semiconductor substrate **222**C: Multi-layer wiring layer **222**CB: Wiring **222**ED: Dummy electrode **302**: Pixels **302**A: Normal pixel **302**B: Charge ejection pixel **309**: Vertical signal line **312**: Semiconductor substrate **321**: Photoelectric conversion section **322**: Capacitive element **323**: Reset transistor **324**: Amplifying transistor **325**: Selection transistor **341**: Semiconductor thin film **342**: P-type layer **343**, **344**: N-type layer **345**: Anti-reflection film **346**: Color filter **347**: On-chip lens **351**: Passivation layer **352**: Insulating layer **353**A, **353**B: Connection electrode **354**: Bump electrode

Claims

1. A sensor apparatus, comprising: a package substrate that includes: a first recess portion on a first side of a first surface of the package substrate; a second recess portion on a bottom surface of the first recess portion; and a plurality of terminals on a second side of a second surface of the package substrate, wherein the second side is opposite to the first side; a Peltier element in the second recess portion; a circuit substrate on an opposite side of a bottom surface of the second recess portion, wherein the Peltier element is between a bottom surface of the circuit substrate and the bottom surface of the second recess portion, the bottom surface of the circuit substrate is attached to the bottom surface of the first recess portion, and the bottom surface of the circuit substrate faces the Peltier element; and an image sensor on an upper surface of the circuit substrate, wherein the circuit substrate is different from the image sensor, and the upper surface of the circuit substrate is opposite to the bottom surface of the circuit substrate.
2. The sensor apparatus according to claim 1, further comprising: a lid attached to the package substrate; and a sealed airtight space between the lid and the package substrate, wherein the lid is configured to cover the circuit substrate and the image sensor.
3. The sensor apparatus according to claim 2, wherein the lid is seam welded to the package substrate.
4. The sensor apparatus according to claim 1, wherein the circuit substrate and an inner side surface of the second recess portion separate a space around the Peltier element from a space around the image sensor.
5. The sensor apparatus according to claim 1, further comprising a casing in the second recess portion, wherein the casing accommodates the Peltier element, and the casing separates a space around the Peltier element from a space around the image sensor.

6. The sensor apparatus according to claim 5, wherein the casing includes a metal.
7. The sensor apparatus according to claim 1, further comprising a wall member between the circuit substrate and the bottom surface of the second recess portion, wherein the wall member laterally surrounds the Peltier element, and the circuit substrate and the wall member separate a space around the Peltier element from a space around the image sensor.
8. The sensor apparatus according to claim 1, wherein the bottom surface of the circuit substrate includes a conductor layer, and the conductor layer surrounds a region opposed to the Peltier element.
9. The sensor apparatus according to claim 1, wherein the image sensor has a photoelectric conversion layer that includes silicon.
10. The sensor apparatus according to claim 1, wherein the image sensor has a photoelectric conversion layer that includes a compound semiconductor.